

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

Key relationships:

- Power dissipated in a resistor: $P = I^2 R = V^2 / R$, with $P = 40\text{ W}$ and $V = 12\text{ V}$.
- Current via a resistor: $I = V / R$.

Solve for resistance (two equivalent ways):

- Using $P = V^2 / R$: $R = V^2 / P = 144 / 40 = 3.6\text{ }\Omega$.
- Using $I = V / R$ and $I^2 R = P$: $R = P / I^2 \hat{=} I = \text{sqrt}(P/R) = \text{sqrt}(40/3.6) \hat{=} 3.33\text{ A}$; checking $I \hat{\cdot} R = 3.33 \hat{\cdot} 3.6 \hat{=} 12\text{ V}$, consistent.

Convert:

$I = 3.33\text{ A}$, $R = 3.6\text{ }\Omega$.

Correct Answer: E. 3.33 amps, 3.6 ohms

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<answer>

E. 3.